APEX TUITIONS

Daily Worksheet Series

Grade 8 • Science

CBSE / State Board Aligned

Student Name: _____ **Roll No / Sec:** _____ **Date:** _____

Max Marks: 25 **Time Allowed:** 40 Mins **Total Questions:** 11

DAILY PRACTICE WORKSHEET
Class: 8 | **Subject:** Science
Topic: Force and Pressure | **Curriculum:** CBSE / State Board

Name: _____ **Roll No.:** _____ **Date:** _____

SECTION A: Conceptual Recall / Objective

(Answer all questions in the space provided)

Q1. The SI unit of force is _____, and the SI unit of pressure is _____.

Q2. Which of the following is an example of a non-contact force?
(a) Muscular force
(b) Frictional force
(c) Electrostatic force
(d) Tension force
Answer: [_____]

Q3. State True or False: Atmospheric pressure increases as we move higher above sea level.
Answer: _____

Q4. What type of force is responsible for the downward motion of a fruit falling from a tree?
Answer: _____

Q5. Fill in the blank: Liquid pressure at a given depth acts equally in _____ directions.

SECTION B: Short Practice / Working Problems

(Show necessary steps and working where applicable)

Q6. A wooden block applies a perpendicular force of 300 N on a flat surface having an area of 6 m^2. Calculate the pressure exerted by the block on the surface.
\ \ \ \ **Q7.** State two points of difference between contact forces and non-contact forces, giving one example of each.
\ \ \ \ **Q8.** A sharp knife cuts vegetables more easily than a blunt knife, even when the same force is applied. Explain the reason using the relationship between pressure, force, and area.
\ \ \ \ **Q9.** A metal cube of weight 60 N has a base edge of 0.5 m. Calculate the pressure it exerts on the ground.
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SECTION C: Application & Critical Thinking

Q10. Porters place a round piece of cloth on their heads when they have to carry heavy loads.
(a) Explain the scientific principle behind this practice.
(b) What happens to the pressure felt on the porter's head if the contact area is doubled while the load remains constant?
\ \ \ \ \ **Q11.** Take a tall plastic bottle with three identical pinholes made along its vertical side: one near the top, one in the middle, and one near the base. The bottle is filled completely with water.
(a) From which hole will the stream of water shoot out the farthest? Justify your answer.
(b) What conclusion can be drawn from this observation regarding pressure exerted by liquids?
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